PARSHVANATH CHARITABLE TRUST'S

A. P. SHAH INSTITUTE OF TECHNOLOGY

(All Branches NBA Accredited)

Department of Information Technology

Academic Year: 2025-26
Semester: IV
Class / Branch: SE- IT A
Subject:Network design lab
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Roll No.69
Date of Submission:08/04/2026

EXPERIMENT NO 11

AIM: To design and implement connection-oriented network application through socket API

OUTPUT:

1) Server start

```
nagios@worker01:/home/kbd/socket$ ./tcps
Server is waiting...
Connected Client IP: 192.168.104.26
Connected Client Port: 39624
Received number: 5
Result sent: 120
```

2) Client start

```
nagios@worker01:/home/kbd/socket$ cc -o tcpc tcpc.c
nagios@worker01:/home/kbd/socket$ ./tcpc
Enter a number: 5
Factorial = 120
nagios@worker01:/home/kbd/socket$
```

3) syn flag

```
09:08:24.617082 IP 192.168.104.26.39624 > 192.168.104.26.8000: Flags [S], seq 42
81404945, win 65495, options [mss 65495,sackOK,TS val 3676487622 ecr 0,nop,wscal
e 7], length 0
        0x0000:  4500 003c 3dae 4000 4006 ab88 c0a8 681a  E..<=.@.@.....h.
        0x0010:  c0a8 681a 9ac8 1f40 ff31 0e11 0000 0000  ..h....@.1......
        0x0020:  a002 ffd7 51b4 0000 0204 ffd7 0402 080a  ....Q...........
        0x0030:  db22 bfc6 0000 0000 0103 0307            ."..........
09:08:24.617113 IP 192.168.104.26.8000 > 192.168.104.26.39624: Flags [S.], seq 6
14587786, ack 4281404946, win 65483, options [mss 65495,sackOK,TS val 3676487622
 ecr 3676487622,nop,wscale 7], length 0
        0x0000:  4500 003c 0000 4000 4006 e936 c0a8 681a  E..<..@.@..6..h.
        0x0010:  c0a8 681a 1f40 9ac8 24a1 dd8a ff31 0e12  ..h..@..$....1..
        0x0020:  a012 ffcb 51b4 0000 0204 ffd7 0402 080a  ....Q...........
        0x0030:  db22 bfc6 db22 bfc6 0103 0307            ."...".......
09:08:24.617140 IP 192.168.104.26.39624 > 192.168.104.26.8000: Flags [.], ack 1,
```

4) PSH FLAG

```
        0x0000:  4500 0034 3daf 4000 4006 ab8f c0a8 681a  E..4=.@.@.....h.
        0x0010:  c0a8 681a 9ac8 1f40 ff31 0e12 24a1 dd8b  ..h....@.1..$...
        0x0020:  8010 0200 51ac 0000 0101 080a db22 bfc6  ....Q........"..
        0x0030:  db22 bfc6                                ."..
09:08:27.662804 IP 192.168.104.26.39624 > 192.168.104.26.8000: Flags [P.], seq 1
:5, ack 1, win 512, options [nop,nop,TS val 3676490667 ecr 3676487622], length 4
        0x0000:  4500 0038 3db0 4000 4006 ab8a c0a8 681a  E..8=.@.@.....h.
        0x0010:  c0a8 681a 9ac8 1f40 ff31 0e12 24a1 dd8b  ..h....@.1..$...
        0x0020:  8018 0200 51b0 0000 0101 080a db22 cbab  ....Q........"..
        0x0030:  db22 bfc6 0500 0000                      ."......
09:08:27.662830 IP 192.168.104.26.8000 > 192.168.104.26.39624: Flags [.], ack 5,
 win 512, options [nop,nop,TS val 3676490667 ecr 3676490667], length 0
        0x0000:  4500 0034 6942 4000 4006 7ffc c0a8 681a  E..4iB@.@.....h.
        0x0010:  c0a8 681a 1f40 9ac8 24a1 dd8b ff31 0e16  ..h..@..$....1..
        0x0020:  8010 0200 51ac 0000 0101 080a db22 cbab  ....Q........"..
```

4)FYN FLAG

```
09:08:27.662948 IP 192.168.104.26.39624 > 192.168.104.26.8000: Flags [.], ack 9,
 win 512, options [nop,nop,TS val 3676490668 ecr 3676490668], length 0
        0x0000:  4500 0034 3db1 4000 4006 ab8d c0a8 681a  E..4=.@.@.....h.
        0x0010:  c0a8 681a 9ac8 1f40 ff31 0e16 24a1 dd93  ..h....@.1..$...
        0x0020:  8010 0200 51ac 0000 0101 080a db22 cbac  ....Q........"..
        0x0030:  db22 cbac                                ."..
09:08:27.662991 IP 192.168.104.26.8000 > 192.168.104.26.39624: Flags [F.], seq 9
, ack 5, win 512, options [nop,nop,TS val 3676490668 ecr 3676490668], length 0
        0x0000:  4500 0034 6944 4000 4006 7ffa c0a8 681a  E..4iD@.@.....h.
```

5) NETSTAT

```
tcp        0      0 0.0.0.0:8000            0.0.0.0:*               LISTEN
tcp        0      0 0.0.0.0:8000            0.0.0.0:*               LISTEN
tcp        0      0 0.0.0.0:8000            0.0.0.0:*               LISTEN
tcp        0      0 0.0.0.0:8000            0.0.0.0:*               LISTEN
tcp        0      0 192.168.104.26:39624    192.168.104.26:8000     ESTABLISHED
tcp        0      0 192.168.104.26:8000     192.168.104.26:39624    ESTABLISHED
tcp        0      0 0.0.0.0:8000            0.0.0.0:*               LISTEN
tcp        0      0 192.168.104.26:39624    192.168.104.26:8000     ESTABLISHED
tcp        0      0 192.168.104.26:8000     192.168.104.26:39624    ESTABLISHED
tcp        0      0 0.0.0.0:8000            0.0.0.0:*               LISTEN
tcp        0      0 192.168.104.26:39624    192.168.104.26:8000     ESTABLISHED
tcp        0      0 192.168.104.26:8000     192.168.104.26:39624    ESTABLISHED
tcp        0      0 192.168.104.26:39624    192.168.104.26:8000     TIME_WAIT
tcp        0      0 192.168.104.26:8000     192.168.104.26:39624    TIME_WAIT
tcp        0      0 192.168.104.26:39624    192.168.104.26:8000     TIME_WAIT
```

Conclusion

The TCP client-server architecture was successfully implemented using socket programming in C, and reliable communication between client and server was established. Packet analysis using tcpdump confirmed the TCP three-way handshake and demonstrated correct data transmission and protocol behavior.